

Maryam Aminu Mukhtar

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EDUCATION

Northeastern University, Boston, MA

December 2025

Master of Science, Wireless and Network Engineering

Courses: Fundamentals of Computer Networks, Wireless Sensor Networks and the Internet of Things, Mobile and Wireless Networking

University of Massachusetts Boston, Boston, MA

May 2023

Bachelor of Science, Computer Engineering (magna cum laude)

- Awarded Engineering Department's Outstanding Achievement in Engineering Award for exceptional academic and research work

TECHNICAL SKILLS

- **IoT & Connectivity:** Wireless Sensor Networks, Device-to-Cloud Communication, RF Protocols (IEEE 802.15.4, XBee), Arduino, Raspberry Pi, Server-Client Architecture, Real-time Monitoring, Network Troubleshooting, Communication Systems
- **Network & Systems:** TCP/IP, UDP, LAN/WAN, DNP3, 5G NR, Routing Protocols, Wireshark, GNU Radio, ns-3 simulator
- **Languages:** Python (Pandas, NumPy, Matplotlib, Seaborn), C/C++, MATLAB, bash
- **Tools & Platforms:** Mac OS, Linux, Ubuntu, Microsoft Office Suite, GitHub, Docker, Salesforce, NetCloud, Ignition SCADA

SOFT SKILLS: Communication (Written & Verbal), Leadership, Teamwork, Attention to Detail, Problem-Solving, Project Management

PROJECT EXPERIENCE

University of Massachusetts Boston, Boston, MA

Augmented Reality Informational System (ARIS) for Astronauts, Capstone project

Sept 2022 – May 2023

- Designed **end-to-end IoT system** integrating distributed **sensors**, **wireless communication (RF XBee/IEEE 802.15.4)**, edge computing (**Raspberry Pi** server), and cloud connectivity for real-time health monitoring
- Led multidisciplinary team in developing full-stack connected system: sensor nodes (**Arduino**) → **wireless network** → central hub → **data visualization**, demonstrating complete IoT product development lifecycle
- Deployed **machine learning** algorithms for **predictive analytics** and real-time sensor data analysis with **Python-based server-client architecture** for efficient data collection and remote monitoring
- Implemented **bidirectional wireless communication** using Zigbee RF modules and IEEE 802.15.4 protocol for full-duplex data transmission between distributed devices and a central monitoring system

Northeastern University, Boston, MA

5G Network Performance Analysis and Optimization, EECE 7364 Lab Series

Feb 2025 – April 2025

- Conducted hands-on experiments with wireless connectivity **testbed (OAIBOX)**, analyzing factors affecting device communication reliability: bandwidth allocation, signal quality, modulation efficiency, and network congestion
- Diagnosed and resolved connectivity issues through **systematic testing** and analysis of throughput, latency, signal quality (RSRP/RSRQ/RSSI), and packet loss
- Analyzed real-time performance data using **Python (Pandas, NumPy, Matplotlib, Seaborn)** and network analysis tools (Wireshark, iperf) to visualize trends and optimize device-to-network communication across test scenarios

WORK EXPERIENCE

Institute for Intelligent Networked Systems, Northeastern University, Boston, MA

Lab Research Assistant

Sept 2023 – April 2024

- Contributed to projects by deploying **Docker** containers, including **GNU Radio** and web server environments, to advance system efficiency and streamline operations
- Optimized large sensor data for preprocessing initiatives using **Python (NumPy and Pandas)** to accurately prepare datasets, improving data organization for machine learning analysis
- Developed a **web server** using advanced **data visualization** techniques to deliver accessible, real-time data that significantly enhances project outcomes

Sullivan and McLaughlin, Dorchester, MA

Network Operations Co-op

May 2024 – December 2024

- Diagnosed and resolved network connectivity failures across 100+ routers using **NetCloud** management platform, performing real-time monitoring, remote resets, modem diagnostics, and fault isolation to determine hardware vs. configuration root causes
- Processed and configured Cradlepoint LTE/5G routers including SIM coordination, **NetCloud** enrollment, and connectivity validation for enterprise client orders using **Salesforce**
- Coordinated directly with carriers and vendors (Cradlepoint) on the **RMA process** for unrecoverable devices and on SIM/activation issues, managing replacements to minimize client disruption

LEADERSHIP & INVOLVEMENT

Paper Trails & Tales Book Club, Founder & Creative Director, Boston, MA

May 2024 – Present

- Founded and manage a book club, coordinating monthly discussions, curating diverse locations and activities for 18+ members
- Design all promotional materials, manage social media presence, and create engaging visual content to drive member engagement and community growth